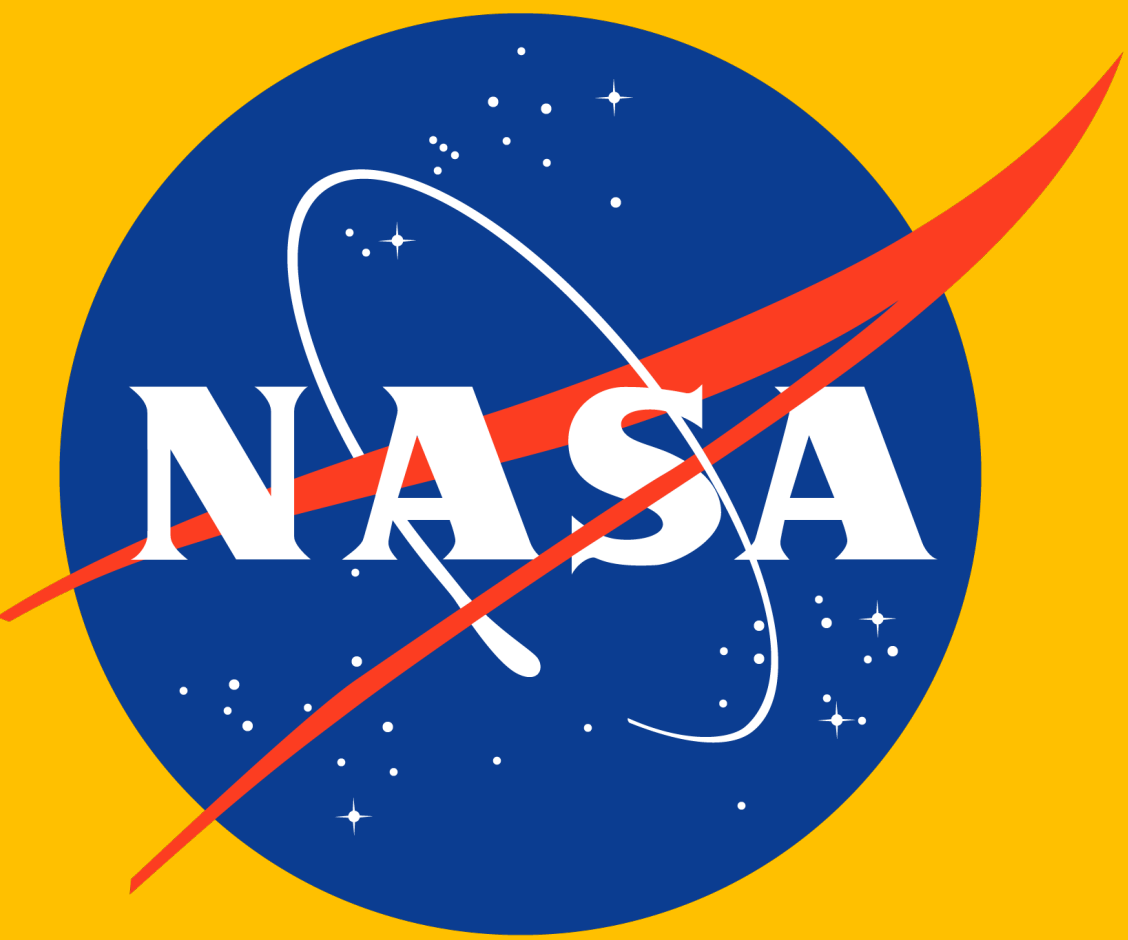




Development of a Thermal -Vacuum Chamber for Extending Calibration of Optical Particle Detectors to Stratospheric Conditions

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Background

Hypersonic flight dynamics are complex and certain important issues, such as what can trigger laminar hypersonic flows to become turbulent, are not well understood; thus new advances in science need to be made before aircraft can be designed that can safely and routinely fly at many times the speed of sound [1]. A team of undergraduate and graduate student researchers in the Aerospace Engineering and Mechanics (AEM) Department, working on a recently funded MURI grant [2], are using optical particle detectors flown on weather balloon missions to characterize the particulate content of the stratosphere. Data collected will be used in computer simulations that study the onset of turbulence in hypersonic flows.

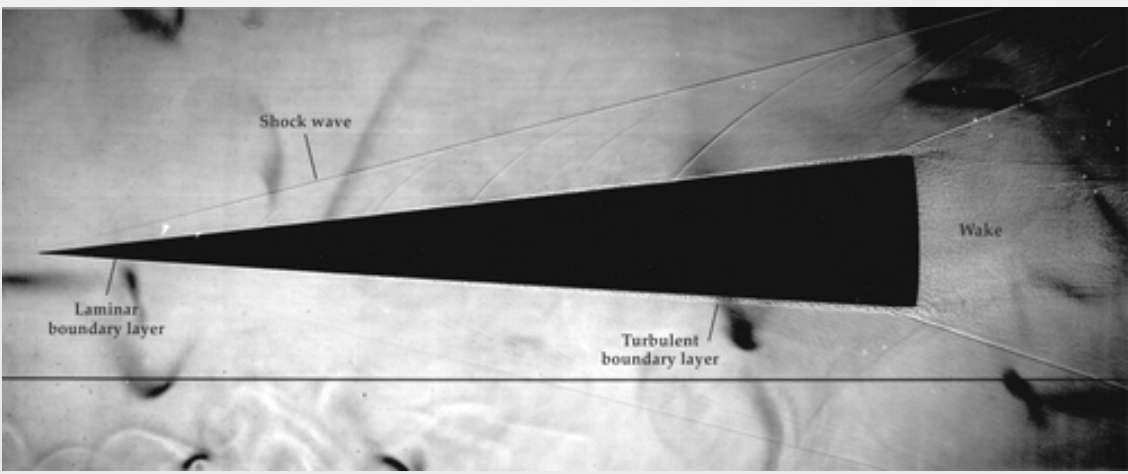


Figure 1 : A shock wave emanating from the nose of a cone travelling at Mach 4 in a ballistic range [from Ref. 1].

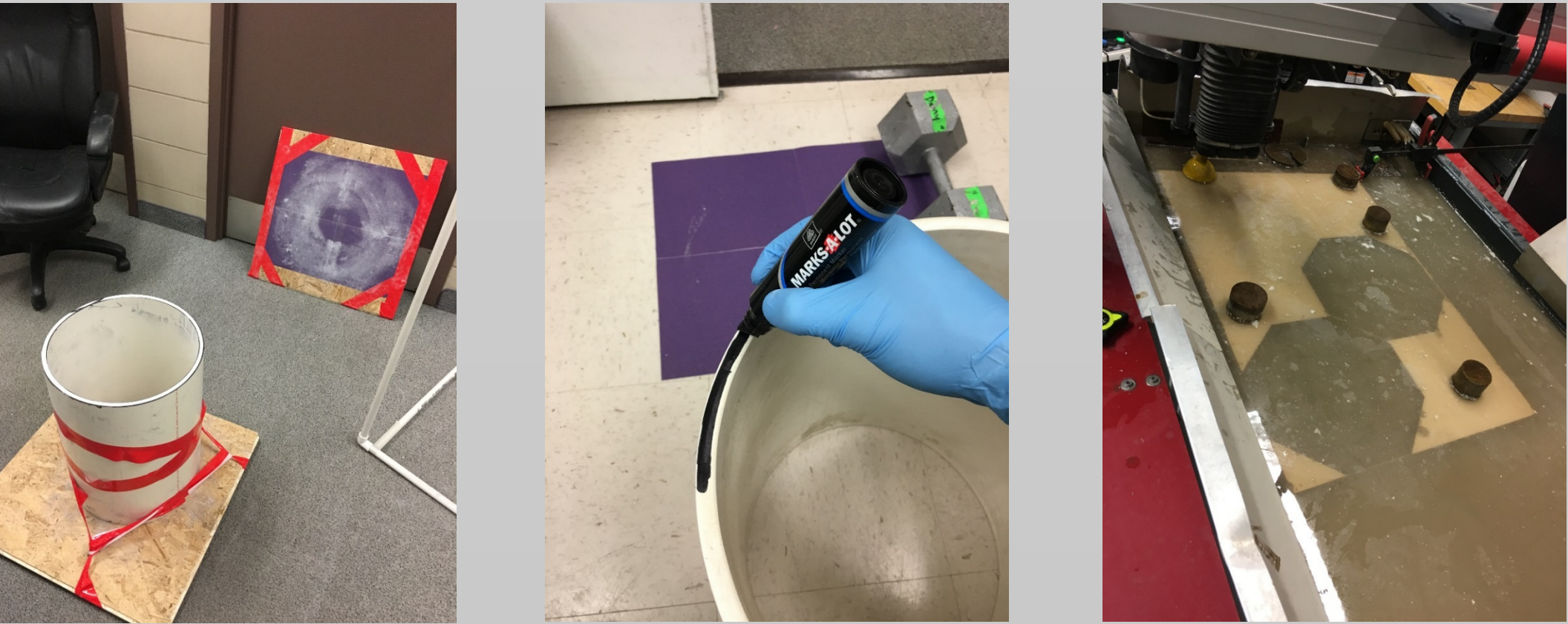
Introduction

The purpose of this project was to build a low -cost thermal -vacuum chamber that will be used to calibrate optical particle detectors such as the modest -cost Alphasense OPC-N3 (\$500) [3] and low -cost Plantower PMS-5003 (\$50) [4] against more -advanced optical particle counters such as the high-cost LOAC detector (\$15,000) [5].

Secondary objectives for the project include 1) testing other payloads built for stratospheric balloon flights and 2) testing payloads from other teams, such as the SmallSat team's SOCRATES CubeSat which will be launched into orbit in late 2019. Testing these payloads in such a chamber can help identify and mitigate potential failures associated with exposure to extreme environmental conditions .

Methods

Our chamber was designed and built based on a design by Howard Brooks of DePauw University [6]. We adopted parts of Brooks' design then improved upon it, allowing us to reach lower temperatures and lower pressures. Lower pressures correspond to conditions at higher altitudes in the stratosphere, of interest to the MURI research project.



Figures 2 -4: Methods of sanding the PVC pipe and cutting out the Lexan endplates with a waterjet cutter



Figure 5: The Arduino Mega computer system in the chamber for monitoring temperature and pressure conditions.



Figures 6 -8: The three length-configurations of the chamber.

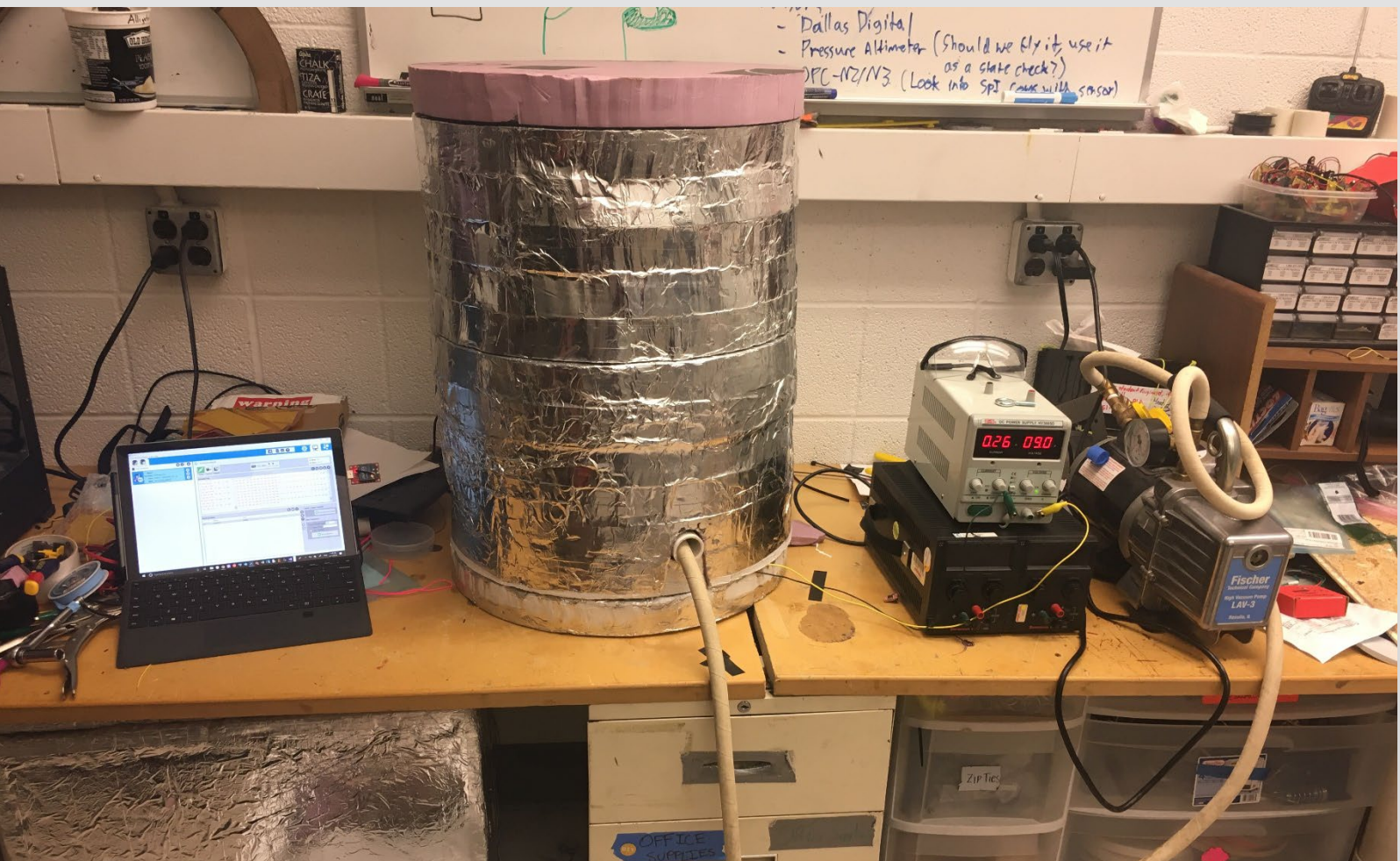


Figure 9: Setup of a thermal-vacuum test with vacuum pump (right), power supply, insulation, and XBee radio.

Notable Payloads Tested

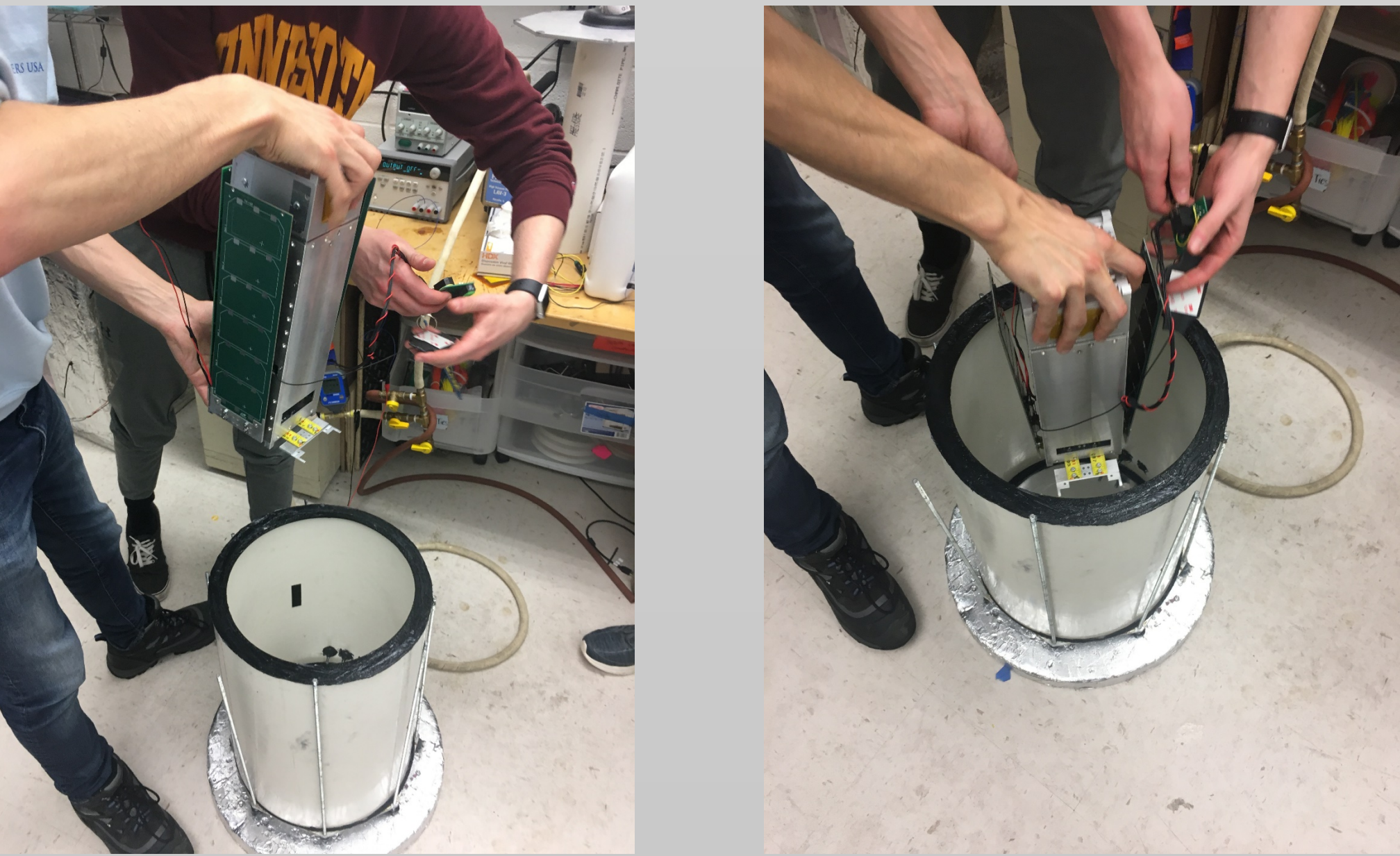
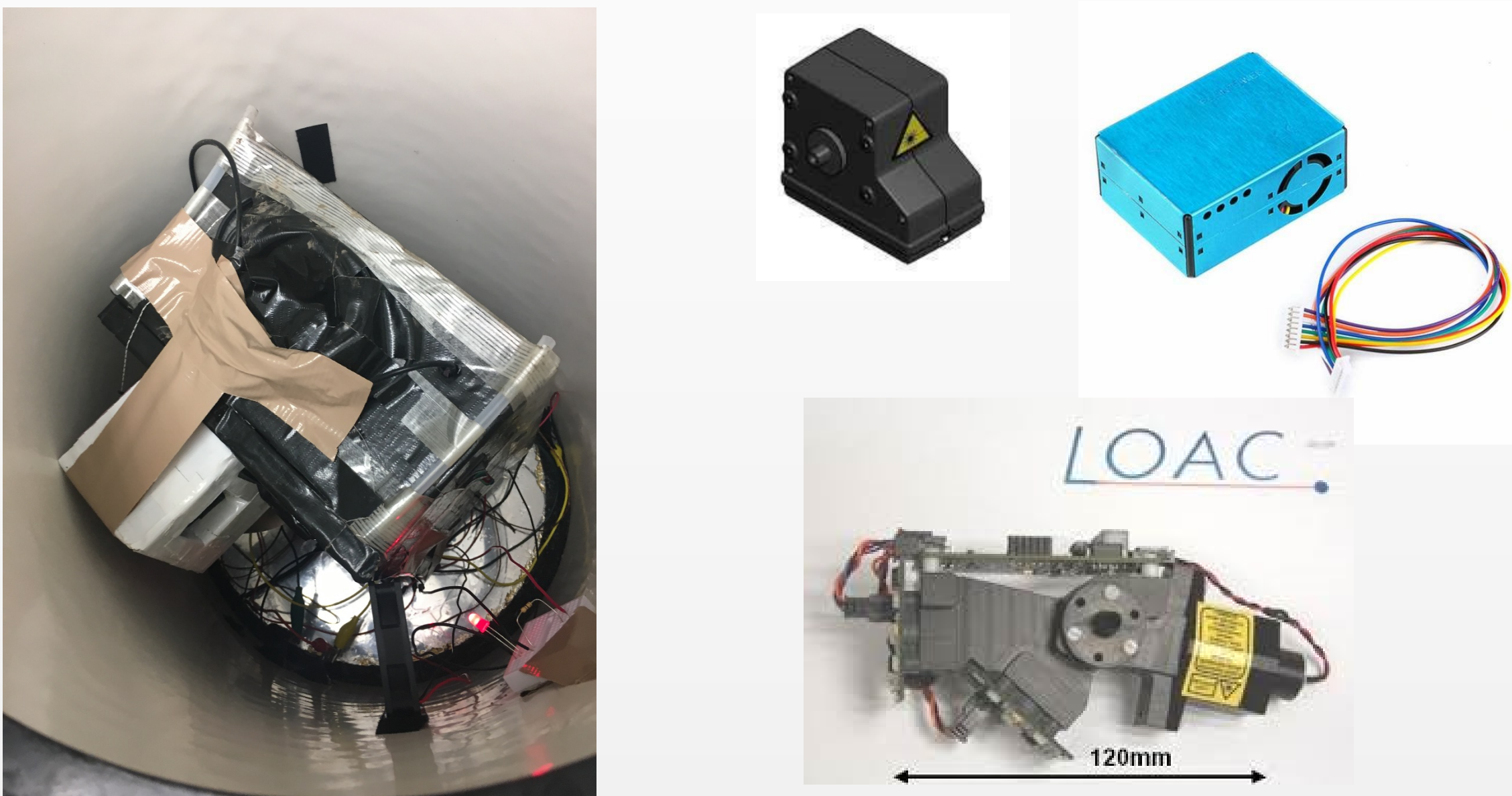


Figure 10: The SOCRATES CubeSat being placed in the thermal-vac chamber for a low-temperature/low-pressure test on April 11, 2019.



Figures 11 -14: A payload containing an Alphasense and a Plantower particle detector being thermal-vac tested on April 2, 2019. Some Alpha-sense OPC-N3 results are plotted in Fig. 18. LOAC detector tests are coming soon. [Images from Ref. 3, 4, 5]

Results

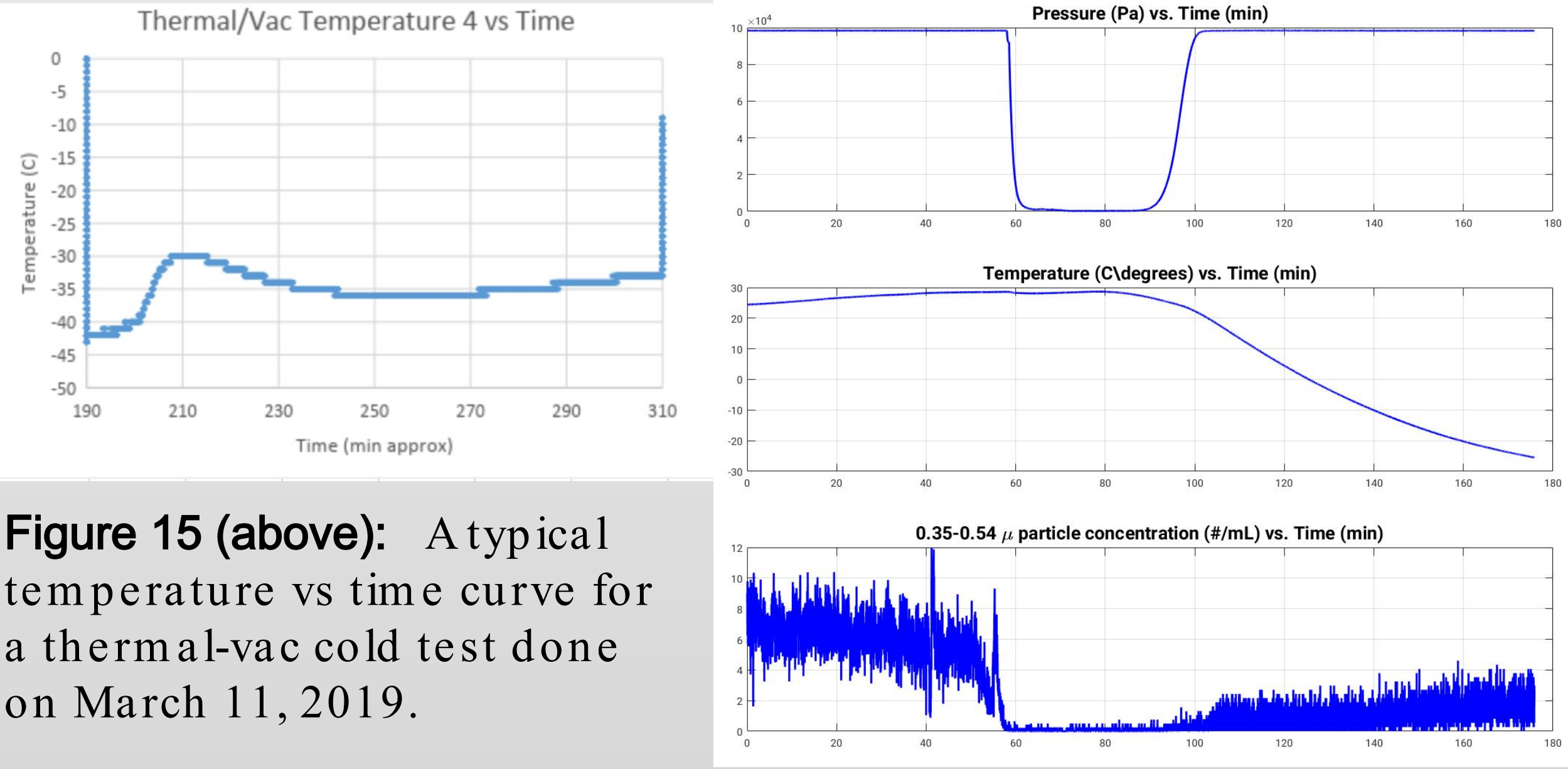


Figure 15 (above): A typical temperature vs time curve for a thermal-vac cold test done on March 11, 2019.

Figures 16 -18 (above, right): Particle data taken by the 0.35 to 0.54 micrometer channel from an Alphasense OPC-N3 particle detector during the April 2, 2019 test pictured above in Figure 11.

Future Work

When we start using LOAC detectors in the near future, we plan to perform tests in the environment of the chamber with all three detector types present simultaneously. In collaboration with Dr. Chris Hogan and AEM graduate student Joseph Habeck, a second thermal-vac system capable of introducing and controlling known amounts of specific-size particles will also be developed for more-precise particle detector calibration. Functionality of the current chamber may evolve farther still and it will continue to be useful for ground testing of additional stratospheric and CubeSat (outer space) payloads.

Acknowledgements

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4. Plantower PM2.5 PMS 5003, <http://plantower.com/en/content/2108.html>
5. J. B. Renard et al, LOAC: a small aerosol optical counter/sizer for ground-based and balloon measurements of the size distribution and nature of atmospheric particles – Part 1: Principle of measurements and instrument evaluation, Atmos. Meas. Tech. Discuss., 8, 9993–10056, 2015
6. Homebuilt High Altitude Test Chamber, Howard Brooks, Academic High Altitude Conference 2014 plus supplementary materials at <https://www.depauw.edu/academics/departments-programs/physics-astronomy/departmentsresearch/base/>.

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